



NewBear Computing Store Ltd



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77-68 Sixth Newsletter July 1979

Comment

Well here we are again. You may wonder how often the newsletter is published. Regretfully the answer is when it is published. It generally takes two months from my first scratchings to a finished product and another month to get it distributed. Often contributions take quite some time to knock into shape particularly when program listings and diagrams are involved. On the topic of listings I have found it far too difficult to take handproduced programs and assemble them. Unfortunately programs are often written in very sloppy ways and the text rarely corresponds to the binary. Thus in future, and in keeping with other publishers, I can only really publish programs which are supplied in a suitable form for assembly, that is they correspond to the HEX. Alternatively S dumps are acceptable if you want to keep your code secret. This does not apply to patches to other programs such as BASIC systems.

This time software is again to the fore.

T-BUG is now available and, in fact, I have had a copy for some time. It is ideal for a 77/68 with a MON 2 board and a memory map VDU. Contrary to early documents the MON 2 single step feature operates correctly. In addition terminals with slow carriage returns are correctly dealt with by the inclusion of PAD characters after the carriage return/line feed sequence. The documentation includes a full assembly listing of T-BUG complete with the S format dump. It can, therefore, easily be modified to run on MON 1 or other hardware. In fact NEWBEAR has had it running on an EPROM board.

At last the famed 32K dynamic RAM board is available as advertised and described in the ACC newsletter. Readers may well have been puzzled as to what those interesting rectangles were- puzzle no more.

X12/13/4/5	74LS153	X2/22/11	74LS74
X23-38	4116	X19	74LS32
X20/21	DM81LS97	X14/6/7/15	74LS163
X9	74LS139	X17	74LS30
X1/8/16	74LS04	X18	74LS00

I have been attempting to down-line load my 77/68 from a large computer over an acoustic coupler. Success is mixed. The principle problem is that the asynchronous clock on the MON 2 board and the CUTs board are not very accurate and moreover drift alarmingly with voltage. The difficulty is that although the

clocks are good enough for single character transmission when long strings of characters are received there is little slack at the end of a character to resynchronise on the next start bit. The first improvement is, of course, to clock the MON 2 serial input from the CUTs board on reading and to clock the CUTs from MON 2 on writing. The second improvement is to derive the write timing from a crystal source (such as the dynamic RAM board). I can assure you that the improvements are well worth while. My belief is that MON 2 should be redesigned to provide a crystal clock. Perhaps one of our readers will oblige? If you do oblige an accurate layout diagram is essential since it takes far longer to layout and track a board than to do the logic design.

Terry Cassell and myself have been mounting software under T-BUG which was written to run under MIKBUG. We have had minimal problems getting it to work but each program has needed individual attention- if any. Terry has a keyboard on his PIA and I use a teletype on the ACIA port thus we develop different code. In particular the break routines in BASIC differ.

TINY BASIC is easy to change, only the break routine at 078D needs changing and the documentation is helpful. SWTP 4K BASIC is also easy to get going with the break routine at 0284. SWTP 8K BASIC is more complicated. This product does not use the MIKBUG I/O routines but does assume MIKBUG hardware. In fact the MIKBUG I/O code is embedded in the BASIC around 038E. The jump to the input routine is at 0356 which should be changed to 7EE1D1. The jump to the output routine is at 0365 which should be 7EE1AC. The break routine is at 0373. Unfortunately a load of PADs are output in front of each line which makes a mess of the output on the screen if a VDU board is in use. The cure is to replace 0B0D to 0B40 and 0AEE to 0AF7 with no-ops. 1103 should be 20, 0429 to 0431 should be 00 and 01CE should be 00. We leave the break routine as an exercise to the reader (you cannot have it all on a plate). All numbers above are in HEX.

All the other programs needed less attention. In fact Terry has checked all the programs Tim sells to ensure they work under T-BUG.

Many of the software problems encountered by users have been caused by the use of MON 1. Most software is designed to run under MIKBUG and like monitors and code thus requires considerable changes to run under the various MON 1 monitors. There appears to me to be only one sensible solution and that is to modify the MON 1 board to have the same memory map as MON 2 thus allowing MIKBUG or T-BUG to be loaded. Having done this most MIKBUG software should run unaltered (apart from break routines possibly). How about someone producing the mods?

Thank you for your many letters and I am sorry that lack of space and time prevent me publishing more of them.

Paul Bryant
The Bumbles
Well Meadow
Shaw
Newbury.

Correction To Design Note 53 DYNARAM

Number pages with front cover as 1.

Page 1 line 5 after 'provides' add 'a clock generator for'.

Page 1 omit lines 7 and 8 and add 'The following signals are used to interface the memory board to the CPU board and are fully compatible with the 77/68 bus structure.'

Page 1 line 17 replace '16 bit' by '64K byte'.

Page 1 line 23 replace 'no special purpose chips are required' by 'by standard TTL LS chips'.

Page 2 line 18 add ',by bringing any one of these pins low the memory board will be deselected'.

Page 2 line 20 replace 'protected' by 'write protected' and add 'Pin 2 on diagram'.

Page 2 foot of page add 'Switch position 1 is furthest away from the edge connector'.

Page 3 line 2 delete 'MB' and '(NE or H)'

Page 3 line 4 add 'or DM81LS95'

Page 3 line 11 add 'diode'.

Page 3 line 13 after 'R8' add ',R9,R10' and omit '(TR4)'.

Page 3 line 14 add .

Page 3 line 15 replace 'X18' by 'X17'.

Page 3 line 17 replace 'X10' by 'SW1'.

Page 3 add at foot of page

TYPICAL POWER CONSUMPTION

+12V @ 0.6A

-12V @ 0.1A

+5V @ 0.5A

These are maximum figures and are typically much less.

Page 6 line 2 add 'In fact it is essential that VBB is never more positive than VSS according to MOSTEK'.

Page 6 line 7 add ', Pin 73.'

Page 6 line 14 add 'x=do not care state,can be 1 or 0'

Page 6 bottom line replace 'programme' by 'program'..bit pedantic!

Page 7 line 8 replace 'minimum' by 'a minimum amount of'.

Page 7 line 10 replace 'glass' by 'magnifying glass'.

The schematic diagram

Edge connector 10 should be labeled CLK.

The -5V should be shown going to edge connector 73.

The layout diagram has lost the vast majority of the hole markers. The wire 4 above X26 should continue to the left. X10 should extend further to the left.
The board has a number of holes absent.

Question

..have any errors been reported on issue 1 PCB's yet (for the dynaram)?

The design note (53) for the Dynaram PCB is not up to your usual high standard....

There was no test program....

My system consists of CPU, VDU, MON1, 3x4K RAM, PIOA, ROM-A, DN35 Prom programmer and 3 prototype boards. I am very pleased with the 77/68 system.

Alan Galpin

Ed. And so say all of us.

Correction to Design Note 47 ROM-A

In fig. 2 the link points are incorrectly labled. For j' read g' for m' read j' for p' read m' for s' read p' for j read g for m read j for p read m for s read p.

Correction To MON 2 board Design Note 38

please number pages with 1 as the front cover.

Page 4 line 4 replace co e by code

Page 6 line 21 replace CE6 by CE6

Page 6 line 23 replace CE9 by CE9

Page 6 line 26 & 29 replace RESET by RESET

Page 7 line 8 replace A12 by A12 and A11 by A11

Page 7 line 14 replace RT1 by RT1

Page 7 line 22 replace CE7 by CE7

Page 10 line 13 & 15 replace CE6 by CE6

Page 10 line 26 & 28 replace CE9 by CE9

Page 16 line 3 replace BA4 by BA4

Figure 2 line 3 replace CE1 by CE1

Figure 2 line 8 replace CE6 by CE6

Figure 2 line 9 replace CE7 by CE7

Figure 2 line 11 replace CE9 by CE9

Figure 2 line 11 replace 111100100 by 1111001

Figure 2 line 12 replace CE10 by CE10

ROM MONITOR PIN OUTS pin 37 is an input even though it is 0v

ROM MONITOR PIN OUTS pin 66 should be RTS
 ROM MONITOR PIN OUTS pin 70 should be CTS
 ROM MONITOR PIN OUTS pin 73 should be RESET
 ROM MONITOR PIN OUTS for 77-68 read 77-78
 In the layout diagram ----- is 8 way ribbon cable.
 The link BA9-15 allows top half of ROM to be accessed.
 The link 15-0v does not allow the top half of ROM to be accessed.
 Page 16, the dotted box should include 'INSTEAD OF X24'
 The diagram on page 16 shows how to externally clock the serial
 port.
 Parts list, for 74LS127 read 74LS27

Correction To BUG 2 Program

This program produces 32 characters to the line on the VDU. This
 can be changed to 40 characters with the following changes:-

Location FF8C	8C93C0
FF91	CE8FD8
FF95	A650
FF97	A728
FF9A	8C9398

Geoff Haydock

TBUG In The Field.

What a strange place to put a PROM! Here's a list of various
 commercial software that I have tried with TBUG, if any patches
 were required then these are shown. These patches may also be of
 interest to those people trying to implement this software on other
 monitors.

TSC/SWTPC GAMES.

START-TREK, MASTERMIND, CRAPS, CHOMP, ACEY-DUCEY, KLINGON-CAPTURE,
 ROVER, HURKLE, SWITCH and STOCKMARKET. These all loaded and ran
 straight away without modification.

Utilities.

MSI disassembler, BYTE CONSERVER (TINY assembler as in PCW), Jack
 Emmerich's Tiny Assembler (as featured in BYTE, a reprint of the
 relevant articles is available together with a paper tape of the
 object code).

SEARCH (as in PCW), SOFT FACADE (as in PCW) - highly recommended as a
 method of implementing single step/trace, much better than the
 facility provided on the MON-2 board although a prom with the
 relevant software in would be required as the program occupies

about 256 bytes, so it is best to put it in a PROM. Ed Smith's relocating Linking Assembler- also highly recommended, the patches for this are minimal and a list of them is available from the Bear.

BASICS.

SWT 4K BASIC as outlined by Paul-needs 8K RAM and runs nicely.

SWT 8K BASIC just runs in 8K. these patches are for version 1 runs nicely but is rather slow and primitive. But if you must use it then here are the patches.

ADDRESS	DATA
01CE	00
0356	7E E1 D1
0365	7E E1 AC
0429-0431	00'S
0AEF-0AF7	01'S
0B0D-0B40	01'S

If the keyboard is on the A side of the PIA then...

0373	B6 80 11 2A 10 B6 80 10 01 01 01 01 01 01
------	---

If the keyboard is on the B side of the PIA then...

0373	B6 80 13 2A 10 B6 80 12 01 01 01 01 01 01
------	---

CSS BASIC VERSION 4 -needs 12K. A nice BASIC but unfortunately has some bugs. TLOAD does not appear to search for a particular named file as it should but insted loads the first one it comes across. it also reckons that -372 is -9. Here are the patches for TBUG.

ADDRESS	DATA
01F1	7E E1 D1
01FD	7E E1 AC

Before running CSS BASIC for the first time it is best to clear locations 0000-00FF. Here is a routine to replace the break routine with it allows a break from either of the three TBUG ports and so it does not matter how you have configured your keyboard teletype or whatever.

00001		NAM	BREAK
00002	*		
00003	*	SUBROUTINE PERFORMS BREAK WITH CSS	
00004	*	BASIC A BREAK IS ACCEPTED FROM EITHER	
00005	*	OF TBUGS THREE PORTS THE BREAK CHAR	
00006	*	IS CONTROL C.	
00007	*		
00008 0207		ORG	\$0207

```

00009 0207 37      BREAK PSH B
00010 0208 36      PSH A
00011 0209 B6 8000  LDA A  ACIACN  BREAK FROM SER. PORT ?
00012 020C 44      LSR A
00013 020D 25 0A    BCS  BREAK4
00014 020F B6 8011  LDA A  PIAACN  FROM PAR. PORT A ?
00015 0212 2B 05    BMI  BREAK4
00016 0214 B6 8013  LDA A  PIABCN  FROM PAR. PORT B ?
00017 0217 2A 0D    BPL  BREAK8
00018 0219 BD E1AC BREAK4 JSR  INPOLL  GET CHAR AND CHECK IF IT
00019 021C 81 03    CMP A  #$03  CONTROL C ?
00020 021E 26 06    BNE  BREAK8
00021 0220 BD 0C60  JSR  PUTPRT
00022 0223 7E 0A87  JMP  READY
00023 0226 32      BREAK8 PUL A      NO BREAK SO RETURN
00024 0227 33      PUL B
00025 0228 39      RTS
00026              *
00027      0A87  READY EQU  $0A87
00028      0C60  PUTPRT EQU $0C60
00029      8000  ACIACN EQU $8000
00030      8011  PIAACN EQU $8011
00031      8013  PIABCN EQU $8013
00032      E1AC  INPOLL EQU $E1AC
00033              *
00034                      END

```

SYMBOL TABLE

```

BREAK 0207  BREAK4 0219  BREAK8 0226  READY 0A87  PUTPRT 0C60
ACIACN 8000  PIAACN 8011  PIABCN 8013  INPOLL E1AC

```

CSS BASIC VERSION 5 (if you can get it) - same facilities as above. 'DELETED' bug has been cured (see previous newsletter) TLOAD works fine but it still seems to think that -3f2 is -9. Maybe the Yanks do their maths differently over there (could account for Skylab). Sorry chaps. Anyway here's what to do if you want to run CSS V5 with TBUG...

```

ADDRESS      DATA
017A         39
01CA         7E E1 D1
01F5         7E E1 AC

```

Here's the break routine again but slightly modified for version 5.

```

00001          NAM      BREAK
00002          *
00003          *      SUBROUTINE PERFORMS BREAK WITH CSS
00004          *      BASIC A BREAK IS ACCEPTED FROM EITHER
00005          *      OF BUGS THREE PORTS THE BREAK CHAR
00006          *      IS CONTROL C.
00007          *
00008 022E          ORG      $022E
00009 022E 37      BREAK   PSH B
00010 022F 36          PSH A
00011 0230 B6 8000     LDA A  ACIACN  BREAK FROM SER. PORT ?
00012 0233 44          LSR A
00013 0234 25 0A      BCS      BREAK4
00014 0236 B6 8011     LDA A  PIAACN  FROM PAR. PORT A ?
00015 0239 2B 05      BMI      BREAK4
00016 023B B6 8013     LDA A  PIABCN  FROM PAR. PORT B ?
00017 023E 2A 0A      BPL      BREAK8
00018 0240 BD E1AC     JSR      INPOLL  GET CHAR AND CHECK IF IT
00019 0243 81 03      CMP A  #$03  CONTROL C ?
00020 0245 26 03      BNE      BREAK8
00021 0247 7E 0AF9     JMP      READY
00022 024A 32          BREAK8 PUL A      NO BREAK SO RETURN
00023 024B 33          PUL B
00024 024C 39          RTS
00025          *
00026 0AF9  READY EQU  $0AF9
00027 8000  ACIACN EQU  $8000
00028 8011  PIAACN EQU  $8011
00029 8013  PIABCN EQU  $8013
00030 E1AC  INPOLL EQU  $E1AC
00031          *
00032          END

```

SYMBOL TABLE

```

BREAK 022E  BREAK4 0240  BREAK8 024A  READY 0AF9  ACIACN 8000
PIAACN 8011  PIABCN 8013  INPOLL E1AC

```

Terry Cassell

8x8 Boards

Apparantly Vero have discontinued their plain boards with edge connectors ready plated. A shame since Peter Dunkley has foils for the ETI cassette interface to suit 77/68 board format. It may

be possible to make these available if there is demand.

Op-Codes

R. Clayton has found Op-Code 14 which ANDs Acc A and Acc B. In fact there are a large number of useful(?) op-codes not published and these were listed in a recent Kilobaud (or was it Byte). You use them at your risk since Motorola may well change their masks with development and thus the codes may change.

House Alarm System

Clive Feather has produced some ideas on using 77/68 for controlling various devices round the house. His article is theoretical and discusses a method of running a 'bus' round the house to which many serial devices could be interfaced. It tackles the problem of simultaneous use of the bus by several devices. The article was rather long for the newsletter and the diagrams not easy to reproduce. Please contact Clive for details if interested.

Keyboard Purchased From Display Electronics

Display Electronics have been advertising coded keyboards in Wireless World, price £18 or so. Having purchased one I have the following information.

Code is parallel 8 bit (non-ASCII) with strobe, also two other bits direct from Alpha and Numeric shift keys, making an effective 10 bits. If two ordinary keys (not shift) are pressed then there is no strobe.

Mechanical layout is conventional QWERTY but with numbers not at the top but on the right hand side sharing some letter keys. Some of the minor keys are black and one of them at each side is double size (like a shift key). The keys are two shot moulded so that legend changing would be difficult. The key assemblies clip into a stainless steel frame over a Hall effect switch permanently soldered into a fibre-glass PCB. The following I.C.s are used, all Motorola.

MC846P, 2xRW10038 and 2xRW10039. Plus a couple of transistors and a handful of other components.

There is a gold edge connector at the top left corner and +5V at about 1/2amp is needed. Some or all of the coding is done on a crossbar system on a second PCB folded under the first on a flexible connector. The through connections are unfortunately rivets which are soldered so that major damage might result if

this coding was to be changed. This second board is badly warped but this dose not seem to matter. Having accepted that I could not readily change the on board coding I decided to use the 10 bit code to address a suitably programmed 2708 and get standard ASCII from this.

The 8 data lines are not TTL levels as they swing from +5 to ground so that they appear to be CMOS. I used 4049 (or 4050) CMOS buffers not forgetting pull down resistors to protect the inputs if the keyboard is unplugged, they feed the 2708 with TTL buffers feeding out either parallel ASCII then feeding a UART etc for serial information. I then went to town and made the whole issue readily 'friggable' with switches insted of hard wired UART programming etc so that different bit rates and code formats can be accomodated easily for experimentation. Obviosly also a change of 2708 enables any code format to be handled.

P.G. Dunkley

77-68 VDU Board

On page 10 of the Fifth 77-68 U.G. Newsletter, L.T. Haynes asks why page 9, 6.d of the design note 36 calls for the system reset button to be held operated. It was assumed that at that stage of testing, system memory would contain only rubbish and, if the reset button were not held operated, the CPU would try to run this random 'program', with possibly strange repercussions on the display!

The problem Mr. Haynes encountered with a spike on the output pin 11 of X1 is strange in that when x11 changes from 1110 to 0001, the 1 to 0 change on X11 pin 12 occurs some tens of nano-seconds before the 0 to 1 change on X11 pin 11 and there should therefore be no 'break' causing a spike on X1 pin 11. I would suspect something strange about X11 or possibly X1. However, in the best traditions of computer engineering, if a 'fix' works, then that is the optimum solution.

Having helped fire up several VDU boards, the following may be of help:-

A wavy edge to the display is due to ripple in the +5V line; X15 is sensitive to this. Cure is usually to use a good IC regulator rather than the simple discrete component type shown in the 77-68 CPU manual.

If you're having trouble with spurious characters or characters changing on the display then try the following test:-

Clear the display then enter a single row of characters along the bottom. Then step this row up the screen by entering Carriage Returns, watching carefully for any changes to the characters in the row; which is usually due to a faulty multiplexer IC (X19,20,22 or 23), or for the appearance of spurious characters in other parts of the display. The latter problem is usually caused by the use of el-cheapo slow memory (X27-34) in place of the types specified.

Mike Lord

Ed. El-cheapo indeed! I used extra el-cheapo- penny a chip-works only in a strong wind- ex equipment- ex NewBear sale. The effect of slow store is a certain amount of 'ghosting' of a flickering nature, that is a character or bits of a character also appear in the next character box due to the chip output being still set as for the previous cycle. With careful selection from the mountain of rubbish the effect can be minimised.

MON 2 and Applications

Peter Green has found a number of faults for which I am indebted. He uses SWATBUG and a PETITEVID. He had problems with picture stability which were cured by running the modulator from a battery. Unlike your worthy editor he found the 1200 bit/sec setting up was not critical. He is currently having problems with a SEED cassette interface.

Peter is interested in Geodesy and Astronomy and needs better accuracy than the 8 significant figures given by most BASICs. Can anyone help with, say, a BASIC with 10 significant figures. His work really needs a good disc system. I understand NewBear may offer one shortly.

VDU Board

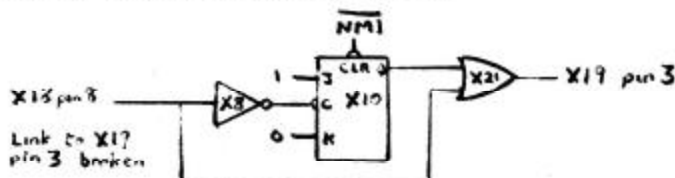
Peter Dunkley has had trouble with X15, the dual 555 timer. R10 had to be changed to 10K to achieve the correct frequency. In addition the 556 appeared to mis-trigger even when as many chips as possible were removed and a separate power supply used as well as extra decoupling. Apparently the 556 is prone to this problem. Peter found that the 4K7 pot RV3 gave insufficient range for the various 556 timers he tried so he used a 2K2 pot which made the setting critical. An aid to setting RV3 is to place the index finger on the leg of RV3 nearest RV2 whence hum can be seen on the picture. RV3 can then be adjusted for minimum beat.

Ed. I had similar problems. I found C2 was sensitive to the type of capacitor used. Peter suggests a change of circuit. Ideally

the line frequency should be derived from a crystal. See page 9 of the 5th nesletter for a suitable circuit.

More VDU Board

Unwanted reads from the keyboard can be suppressed using the circuit below. It requires no new componants.



This mod with suitable code allows easy use of the 40x24 display.

D.A. Roberts

Ed. A suitable Scroll routine was provided but was unfortunately not easily asseambleable. There is a scroll routine which could be used buried in T-BUG.

More BASIC

In CSS BASIC addresses 0125 to 0238 are used for I/O but in addition the code at 1D2A is also used which is:-

```
JSR #E1D1
RTS
JSR #E1AC
RTS
```

This is used for file handling. With non MIKBUG compatible monitors this needs changing.

Geoff Haydock

Computer Ware Super Cassette BASIC requires 9K of store. It assumes a scratch pad at A0xx. Either memory can be put at this address or the CPU 256 Bytes can be used and the BASIC altered. Be careful if your MON 1 monitor uses the CPU storage.

Terry Mitchell

BASIC for Sale

Mick Alger invites you to bring your 77/68 to life with a full feature 8K BASIC. The configuration required is a CPU board, a MON 1 board running BUG 2 and the 77/68 memory mapped VDU board. However the tape he can supply contains a modified version of BUG 2 which enables a printer to be supported through the ACIA B port and enables load and save to take place through ACIA A via CUTS interface. Patches to BUG 1 can be supplied if there is sufficient demand. The instructions available are:-

LIST, RUN, NEW, SAVE, LOAD, WIDTH,
CONT, TRACE ON, TRACE OFF,
REM, DIM, DATA, READ, RESTORE, LET,
FOR, NEXT, STOP, COSUB, END, GOTO,
ON GOTO, ON GOSUB, IF THEN, INPUT,
PRINT, RETURN, POKE, DEF, ABS, INT,
RND, SQN, TAB, PEEK, SIN, COS, TAN,
FNX, POS, LEN, ASC, SQR, EXP, LOG,
VAL, CHRS#, STR#, LEFT#, RIGHT#, MIDS,
ATAN, SCLR.

Available on Kansas City cassette for £45 from

M.D. Alger,
The Old Orchard,
Main Road,
Saltfleetby,
Louth,
Lincs.,
LN11 7SS.

Music?

Try this with the speaker connected in the usual way:-
starting at location 0

96FOXYY4A26FD5CD7FF7A00F2

26F1864097F296F04A97F026E6

ZZ97F020ED

XX and YY are any single byte operations on the A register.

ZZ is anything.

The result is said to be 'therapeutic'. What do you think?

R Clayton

Page Based Text Editor

The editor described takes a page of ASCII coded data from memory (end of page being denoted by an EOT control character HEX04) and puts it on the screen in a 24 line by 32 character format.

Once on the screen the following facilities are available:-

1. Full cursor control ↑ ↓ ← → CR and HOME, this includes full

wrap around cursor movement at the screen edges.

2. Clear page (from cursor), Clear line the cursor is on, clear to the end of the line from the cursor position.
3. Invert printed characters, invert from cursor position, whole screen or characters only.
4. Set CR TAB up to 2 line movement. CLR TAB.
5. Insert and delete characters, insert string between delimiters /..../.
6. Insert and delete line.
7. Copy page to memory at any time during the editing (to address. given) without disturbing the screen contents. This can be used for continuous text editing.
8. Set End of Line 'folding' option - truncated words at end of line are automatically shifted to the start of next line- CLR option.
9. Swap option with 1:1 memory mapped area. provides for a help or command page or facility for editing two pages at once.
10. End Edit; can abort without saving or save to source address of page by CR default, or to a new address to be input.
11. Swap line.

The whole program is 2K bytes including 3/4K bytes for the memory mapped page used by the swap command.

The option can easily be extended to cover other facilities such as left and right justification, centering of headings/lines, exchange strings etc.

A summary of the command options and page start and finish addresses is continuously displayed in the remaining right-hand 8x24 line strip on the VDU.

I have tried to make the program flexible and relocatable as possible by avoiding all unnecessary use of jumps and extended addressing modes, this involves extensive use of direct addressing in the region 0000 to 0020. All the major parts of the editor are available as subroutines and thus the system is readily expandable by anyone with a reasonable knowledge of assembly language. Several monitor routines are used to input 2 and 4 hex numbers for addresses and tabs, printing prompts and clearing the screen initially. Incorporating these would add less than 0080 to the programs size but as most monitors will have these routines readily available it seems unnecessary.

I hope the program will go some way towards filling a chronic software shortage for what is, a very flexible and well designed hardware system. The program as a whole is specifically designed for those systems with relatively limited memory, in that the pages are stored in a minimum size format, and there is also the

advantage of seeing the whole page as opposed to a line editing system.

Robert Laithwaite
Orchard House
Cheshire
WA4 4EN

Ed. Looks good. I think Robert may require some folding stuff for a copy so negotiate with him those who are interested. Alternatively I would like to publish the code, Robert, if you would supply it.

Mastermind

The requirements of this program are a 77/68 CPU board and a MON 1 board running under BUG 1 with a teletype (or similar) terminal.

The program uses OF HEX bytes of the CPU board memory from 0000 so that direct addressing mode can be used. Once loaded the program can be dumped complete with monitor and simply reloaded with 'boot' operation.

The start address is 'FCBB' replying N to the first message will return you to the monitor; Y starts the random number generator and 'any key to start play' stops it. The player then enters a four digit number 0-9 and then carriage return. The program then tests for Hits or Misses and prints the result and the number of goes left. A 'hit' is defined as a correct number in the correct place and a 'miss' as a number correct but in the wrong place. If during a go the player wishes to change a number then insted of the fifth character entered being a carriage return any other character will allow reselection of the complete number without decrementing the go counter.

If after ten goes the correct number has not been tried the the machine considers that it has won.

John Tracey

00001			NAM	MAST
00002	FF90	PNTCH	EQU	\$FF90
00003	F402	ACIA	EQU	\$F402
00004	FF82	GETCH	EQU	\$FF82
00005	F403	ACH	EQU	\$F403
00006	0000		ORG	\$0000
00007	0000 0001	AA	RMB	1
00008	0001 0001	PMKEY	RMB	1
00009	0002 0001	C	RMB	1

00010	0003	0001	D	RMB	1
00011	0004	0001	E	RMB	1
00012	0005	0001	KEY	RMB	1
00013	0006	0002	SAVEX	RMB	2
00014	0008	0002	SAVEX1	RMB	2
00015	000A	0001	F	RMB	1
00016	000B	0002	TPKEY	RMB	2
00017	000D	0002	XREG	RMB	2
00018	FC00			ORG	\$FC00
00019	FC00	0D	PLAY	FCB	\$D,\$A
	FC01	0A			
00020	FC02	50		FCC	/PLAY MASTERMIND Y-N/
	FC03	4C			
	FC04	41			
	FC05	59			
	FC06	20			
	FC07	4D			
	FC08	41			
	FC09	53			
	FC0A	54			
	FC0B	45			
	FC0C	52			
	FC0D	4D			
	FC0E	49			
	FC0F	4E			
	FC10	44			
	FC11	20			
	FC12	59			
	FC13	2D			
	FC14	4E			
00021	FC15	0D		FCB	\$D,\$A
	FC16	0A			
00022	FC17	0D	ANY	FCB	\$D,\$A
	FC18	0A			
00023	FC19	41		FCC	/ANY KEY TO START PLAY/
	FC1A	4E			
	FC1B	59			
	FC1C	20			
	FC1D	4B			
	FC1E	45			
	FC1F	59			
	FC20	20			
	FC21	54			
	FC22	4F			
	FC23	20			
	FC24	53			
	FC25	54			
	FC26	41			
	FC27	52			

	FC28	54			
	FC29	20			
	FC2A	50			
	FC2B	4C			
	FC2C	41			
	FC2D	59			
00024	FC2E	0D	ENT	FCB	\$D,\$A
	FC2F	0A			
00025	FC30	45		FCC	/ENTER 4 DIGIT NO AND CR/
	FC31	4E			
	FC32	54			
	FC33	45			
	FC34	52			
	FC35	20			
	FC36	34			
	FC37	20			
	FC38	44			
	FC39	49			
	FC3A	47			
	FC3B	49			
	FC3C	54			
	FC3D	20			
	FC3E	4E			
	FC3F	4F			
	FC40	20			
	FC41	41			
	FC42	4E			
	FC43	44			
	FC44	20			
	FC45	43			
	FC46	52			
00026	FC47	0D	CR	FCB	\$D,\$A
	FC48	0A			
00027	FC49	2D		FCC	/----/
	FC4A	2D			
	FC4B	2D			
	FC4C	2D			
00028	FC4D	0D		FCB	\$D
00029	FC4E	0D	CRS	FCB	\$D,\$A
	FC4F	0A			
00030	FC50	34	HITSS	FCB	\$34
00031	FC51	48		FCC	/HITS /
	FC52	49			
	FC53	54			
	FC54	53			
	FC55	20			
00032	FC56	30	MISS	FCB	\$30
00033	FC57	4D		FCC	/MISSES /
	FC58	49			

	FC59	53			
	FC5A	53			
	FC5B	45			
	FC5C	53			
	FC5D	20			
00034	FC5E	30	GOES	FCB	\$30
00035	FC5F	47		FCC	/GOES LEFT/
	FC60	4F			
	FC61	45			
	FC62	53			
	FC63	20			
	FC64	4C			
	FC65	45			
	FC66	46			
	FC67	54			
00036	FC68	0D	WIN	FCB	\$D,\$A
	FC69	0A			
00037	FC6A	49		FCC	/I WIN /
	FC6B	20			
	FC6C	57			
	FC6D	49			
	FC6E	4E			
	FC6F	20			
00038	FC70	54	YW	FCC	/THE NUMBER WAS /
	FC71	48			
	FC72	45			
	FC73	20			
	FC74	4E			
	FC75	55			
	FC76	4D			
	FC77	42			
	FC78	45			
	FC79	52			
	FC7A	20			
	FC7B	57			
	FC7C	41			
	FC7D	53			
	FC7E	20			
00039	FC7F	30	WYWI	FCC	/0198/
	FC80	31			
	FC81	39			
	FC82	38			
00040	FC83	20		FCC	/ CORRECT YOU WIN/
	FC84	43			
	FC85	4F			
	FC86	52			
	FC87	52			
	FC88	45			
	FC89	43			

```

FC8A 54
FC8B 20
FC8C 59
FC8D 4F
FC8E 55
FC8F 20
FC90 57
FC91 49
FC92 4E
00041 FC93 00 27 K10K FCB 0,0,0,0,0,0,0,0,0,0
FC94 00 10
FC95 00 03
FC96 00 E8
FC97 00 00
FC98 00 64
FC99 00 00
FC9A 00 0A
FC9B 00 00
FC9C 00 01
00042 FC9D 00 K10KH0 FCB 0
00043 FCA0 ORG $FCA0
00044 FCA0 37 PRST PSH B SAVE NO OF BYTES
00045 FCA1 A6 00 LDA A 0,X FIRST CHARACTER
00046 FCA3 BD FF90 JSR PNTCH JUMP TO PNTCH IN BUG 1
00047 FCA6 08 INX POINT TO NEXT CHAR
00048 FCA7 33 PUL B PULL NO BYTES
00049 FCA8 5A DEC B ONE LESS TO PRINT
00050 FCA9 26 F5 BNE PRST IF NOT LAST BYTE START
00051 FCAB 39 RTS
00052 *KEYRWTO REWRITE KEY TEMP FROM KEY PERM
00053 *HITS DOES NASTY THINGS TO DATA
M0054 FCAC DE 01 KEYRW LDX PMKEY LOADS TWO BYTES FROM PER
00055 FCAE DF 05 STX KEY STORES THEM AT TEMP
00056 FCBO FF FC7F STX WYWI STORE AT I WIN/YOU WIN
00057 FCB3 DE 03 LDX PMKEY+2 LOADS TWO MORE
00058 FCB5 DF 07 STX KEY+2 STORES THEN AND
UWIN9 FCB7 FF FC81 STX WYWI+2 STORES THEM INTO IWIN/YO
00060 FCBA 39 RTS
00061 FCBB CE FC00 START LDX #PLAY INDEX TO START OF STRIN
00062 FCBE C6 17 LDA B #$17 NO OF BYTES
00063 FCC0 BD FCA0 JSR PRST PRINT 'PLAY MASTERMIND
00064 FCC3 BD FF82 GETT JSR GETCH GET Y OR N BUG 1
00065 FCC6 81 59 CMP A #'Y
00066 FCC8 27 07 BEQ PRNY AND GET RANDOM NUMBER
00067 FCCA 81 4E CMP A #'N
00068 FCCC 27 02 BEQ RET RETURN TO MONITOR IF N
00069 FCC2 20 F3 BRA GETT MUST BE RUBBISH GO BACK
00070 FCD0 3F RET SWI RETURN TO MONITOR
00071 FCD1 BD FF90 PRNY JSR PNTCH PRINT Y

```

Linking
Numbers

Add	Code	Label	Operator Operand	Comments
00072	PCD4	CE FC17	LDX #ANY	INDEX TO START OF STRIN
00073	PCD7	C6 17	LDA B #17	NO OF BYTES
00074	PCD9	BD FCA0	JSR PRST	'ANY KEY TO START PLAY'
00075	PCDC	CE 2710 TOPNO	LDX #10000	
00076	PCDF	B6 F403 NEXNO	LDA A ACH	IS ANY CHAR
00077	FCE2	47	ASR A	READY AT ACIA
00078	FCE3	25 05	BCS RED	IF READY OFF TO
00079	FCE5	09	DEX	COUNTING DOWN
00080	FCE6	26 F7	BNE NEXNO	IF NOT ZERO ROUND DOWN
00081	FCE8	20 F2	BRA TOPNO	IS BE ZERO SO LOAD TOP N
00082	FCEA	DF 00 RED	STX 0	POINT TEMP BINARY KEY
00083	FCEC	B6 F402	LDA A ACIA	LOAD FROM ACIA
00084	FCEF	4F	CLR A	KEEP IT IN STEP
00085	FCF0	CE 0000	LDX #0	POINT TO WHERE NO STORED*
00086	FCF3	96 00	LDA A AA	FROM TEMPKEY
00087	FCF5	D6 01	LDA B PMKEY, X	FROM TEMPKEY+1
00088	FCF7	DF 06	STX SAVEX	SAVEX DATA POINTER
00089	FCF9	CE FC93	LDX #K10K	LOADINDEX WITH 1ST CONT
00090	FCFC	7F 0005 CVDEC1	CLR KEY	SAVE A CLEARED
00091	FCFF	E0 01 CVDEC2	SUB B PMKEY, X	TAKE 2ND CONTANT FROM B
00092	FD01	A2 00	SBC A 0, X	TAKE 1ST CONTANT FROM A
00093	FD03	25 05	BCS CVDEC5	OVERFLOW
00094	FD05	7C 0005	INC KEY	
00095	FD08	20 F5	BRA CVDEC2	
00096	FD0A	EB 01 CVDEC5	ADD B PMKEY, X	RESTORE PARTIAL RESULT
00097	FD0C	A9 00	ADC A 0, X	GIVE BACK THE EXTR
00098	FD0E	36	PSH A	SAVE A REG
00099	FD0F	DF 08	STX SAVEX1	
00100	FD11	DE 06	LDX SAVEX	
00101	FD13	96 05	LDA A KEY	
00102	FD15	8B 30	ADD A #30	MAKE ASCII CHAR
00103	FD17	A7 00	STA A 0, X	AND PUT IT BACK
00104	FD19	32	PUL A	RESTORE A
00105	FD1A	08	INX	
00106	FD1B	DF 06	STX SAVEX	SAVEX
00107	FD1D	DE 08	LDX SAVEX1	SAVEX1 BACK TO CONSTANTS
00108	FD1F	08	INX	
00109	FD20	08	INX	
00110	FD21	8C FC9D	CPX #K10KH0	GOT TO THE END
00111	FD24	26 D6	BNE CVDEC1	
00112	FD26	BD FCAC	JSR KEYRW	WRITE FROM KEY PERM
00113	FD29	CE FC2E	LDX #ENT	PRINT ENTER...
00114	FD2C	C6 19	LDA B #19	
00115	FD2E	BD FCA0	JSR PRST	
00116	FD31	86 3A	LDA A #3A	SET UP GO COUNTER
00117	FD33	B7 FC5E	STA A GOES	STORE IN GO COUNTER
00118	FD36	CE FC47 ASP	LDX #CR	
00119	FD39	C6 07	LDA B #7	PRINT
00120	FD3B	BD FCA0	JSR PRST	

00121	FD3E	86 09	LDA A	#9	
00122	FD40	B7 FD4B GET	STA A	DIREC+1	AT DIRECT BYTE
00123	FD43	BD FF82	JSR	GETCH	FROM KEYBOARD
00124	FD46	81 0D	CMP A	#0D	COMPARE TO CR
00125	FD48	27 0F	BEQ	TEST	IF IT IS RETURN
00126	FD4A	97 0D DIREC	STA A	\$0D	AT DIRECT BYTE
00127	FD4C	BD FF90	JSR	PNTCH	PRINT CHARACTER
00128	FD4F	B6 FD4B	LDA A	DIREC+1	FROM DIRECT BYTE
00129	FD52	4C	INC A		DIRECT BYTE +1
00130	FD53	81 0E	CMP A	#0E	IF 4 CHAR THEN RETURN
00131	FD55	27 DF	BEQ	ASP	
00132	FD57	20 E7	BRA	GET	GET ANOTHER CHARACTER
00133	FD59	7A FC5E TEST	DEC	GOES	GO COUNTER DECREMENTED
00134	FD5C	8D 3D	BSR	HITS	BRANCH HITS SUBROUTINE
00135	FD5E	CA 30	ORA B	#30	ASCIIIZE NO OF HITS
00136	FD60	F7 FC50	STA B	HITSS	INTO HITS AREA
00137	FD63	8D 4C	BSR	MISSES	BRANCH TO MISSES
00138	FD65	CA 30	ORA B	#30	ACSTISE NO OF MISSES
00139	FD67	F7 FC56	STA B	MISS	INTI MISSES AREA
00140	FD6A	B6 FC50	LDA A	HITSS	
00141	FD6D	81 34	CMP A	#34	4 HITS MEANS A WIN
00142	FD6F	27 14	BEQ	PRNUW	OFF TO PRINT WIN
00143	FD71	B6 FC5E	LDA A	GOES	
00144	FD74	81 30	CMP A	#30	HAVE WE RUN OUT OF GOES
00145	FD76	27 18	BEQ	PRNIW	PRINT I WIN
00146	FD78	BD FCAC	JSR	KEYRW	
00147	FD7B	CE FC4E	LDX	#CRS	PRINT HITS AND MISSES
00148	FD7E	C6 1C	LDA B	#1C	
00149	FD80	BD FCA0	JSR	PRST	
00150	FD83	20 B1	BRA	ASP	BRANCH FOR ANOTHER GO
00151	FD85	CE FC70 PRNUW	LDX	#YW	
00152	FD88	C6 23	LDA B	#23	PRINT YOU WIN
00153	FD8A	BD FCA0	JSR	PRST	
00154	FD8D	7E FCBB	JMP	START	RETURN TO START
00155	FD90	CE FC68 PRNIW	LDX	#WIN	PRINT I WIN
00156	FD93	C6 1B	LDA B	#1B	
00157	FD95	BD FCA0	JSR	PRST	
00158	FD98	7E FCBB	JMP	START	RETURN TO START
00159	FD9B	5F HITS	CLR B		
00160	FD9C	CE 0005	LDX	#KEY	
00161	FD9F	A6 00 HIT2	LDA A	0,X	
00162	FDA1	A1 04	CMP A	4,X	
00163	FDA3	26 05	BNE	NOHITS	
00164	FDA5	5C	INC B		
00165	FDA6	63 00	COM	0,X	
00166	FDA8	63 04	COM	4,X	
00167	FDA A	08 NOHITS	INX		
00168	FDA B	8C 0009	CPX	#KEY+4	
00169	FDA E	26 EF	BNE	HIT2	

8 8
3 22
3 22
5 82-
0001

0
2
0+2
876

222
872
0001

00001
31
2671
2618

00170	FDB0	39		RTS	
00171	FDB1	5F	MISES	CLR B	
00172	FDB2	CE 0005		LDX	#KEY
00173	FDB5	DF 0D		STX	XREG
00174	FDB7	A6 04	MISS2	LDA A	4,X
00175	FDB9	2B 12		BMI	NGOOD
00176	FDBB	CE 0005		LDX	#KEY
00177	FDBE	A1 00	TEST2	CMP A	0,X
00178	FDC0	26 05		BNE	NOMIS
00179	FDC2	5C		INC B	
00180	FDC3	63 00		COM	0,X
00181	FDC5	20 06		BRA	NGOOD
00182	FDC7	08	NOMIS	INX	
00183	FDC8	8C 0009		CPX	#KEY+4
00184	FDCB	26 F1		BNE	TEST2
00185	FDCD	DE 0D	NGOOD	LDX	XREG
00186	FDCF	08		INX	
00187	FDD0	DF 0D		STX	XREG
00188	FDD2	8C 0009		CPX	#KEY+4
00189	FDD5	26 E0		BNE	MISS2
00190	FDD7	39		RTS	
00191				END	

WE GOT ONE
SO PUT OUT OF MISERY

SYMBOL TABLE

PNTCH	FF90	ACIA	F402	GETCH	FF82	ACH	F403	AA	0000
PMKEY	0001	C	0002	D	0003	E	0004	KEY	0005
SAVEX	0006	SAVEX1	0008	F	000A	TPKEY	000B	XREG	000D
PLAY	FC00	ANY	FC17	ENT	FC2E	CR	FC47	CRS	FC4E
HITSS	FC50	MISS	FC56	GOES	FC5E	WIN	FC68	YW	FC70
WYWI	FC7F	K10K	FC93	K10KHO	FC9D	PRST	FCA0	KEYRW	FCAC
START	FCBB	GETT	FCC3	RET	FCDO	PRNY	FCD1	TOPNO	FCDC
NEXNO	FCDF	RED	FCEA	CVDEC1	FCFC	CVDEC2	FCFF	CVDEC5	FD0A
ASP	FD36	GET	FD40	DIREC	FD4A	TEST	FD59	PRNUW	FD85
PRNIW	FD90	HITS	FD9B	HIT2	FD9F	NOHITS	FDAA	MISES	FDB1
MISS2	FDB7	TEST2	FDBE	NOMIS	FDC7	NGOOD	FDCD		

S00600004B44521B

S113FC000D0A504C4159204D41535445524D494ED3

S113FC104420592D4E0D0A0D0A414E59204B455989

S113FC2020544F20535441525420504C41590D0AF2

S113FC30454E5445522034204449474954204E4FA0

S113FC4020414E442043520D0A2D2D2D2D0D0D0A19

S113FC50344849545320304D495353455320304779

S113FC604F4553204C4546540D0A492057494E20D0

S113FC70544845204E554D4245522057415320305B

S113FC8031393820434F525245435420594F55205F

S111FC9057494E00000000000000000000000074

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S113FCA037A600BDF9008335A26F539DE01DF057B
S113FCB0FFFC7FDE03DF07FFFC8139CEFC00C617A3
S113FCC0BDFCA0BDF8281592707814E270220F386
S113FCD03FBDF90CEFC17C617BDFCA0CE2710B6C3
S113FCE0F4034725050926F720F2DF00B6F4024F96
S113FCF0CE00009600D601DF06CEFC937F0005EQ1F
S113FD0001A20025057C000520F5EB01A90036DFE2
S113FD1008DE0696058B30A7003208DF06DE0808E9
S113FD20088CFC9D26D6BDFCACCEFC2EC619BDFCB1
S113FD30A0863AB7FC5EFCFC47C607BDFCA0860988
S113FD40B7FD4BDDFF82810D270F970DBDF90B608
S113FD50FD4B4C810E27DF20E77AFC5E8D3DCA30D7
S113FD60F7FC508D4CCA30F7FC56B6FC5081342752
S113FD7014B6FC5E81302718BDFCACCEFC4EC61C0C
S113FD80BDFCA020B1CEFC70C623BDFCA07EFCBB94
S113FD90CEFC68C61BDDFCA07EFCBB5FCE0005A6E6
S113FDA000A10426055C63006304088C000926EFA7
S113FDB0395FCE0005DF0DA6042B12CE0005A1008D
S113FDC026055C63002006088C000926F1DE0D0878
S10BFDD0DF0D8C000926E03967
S9030000FC

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MINOTAUR

Terry Cassell has sent me the program below which utilises T-BUG. It is well worth study as an axample of the use of the M6800 cross assembler as well as a nice piece of coding illustrating many useful techniques. Contributors please study.

The MINOTAUR game as shown has been written for TBUG and does not run with MIKBUG, SWTBUG or SMARTBUG ha ha!! It is quite pretty and good fun when run and does a good job of proving that memory mapped VDU's are best! I hope that lots of other people will soon be writing software for TBUG as with that combination all sorts of things are possible. Get to know the PLOT routines and you will soon see how easy it is to implement things as screen editing, light pens etc. I look forward to seeing some listings in the next newsletter.

```

00001          NAM    MINOTAUR
00002 0000      ORG    $0000
00003          *
00004          *-----*
00005          *
00006          * MINOTAUR RULES AS IN P.C.W.
00007          * THIS 6800 VERSION WRITTEN FOR
00008          * 77-68 AND TBUG

```

```

00009      * AUTHOR:= T.M.CASSELL.
00010      * DATE 16-3-79
00011      *
00012      * COORDINATES FOR X AND Y ARE HELD IN ONE
00013      * BYTE MS & BITS HOLDS X COORD 0=>9 LS 4
00014      * BITS HOLDS Y COORD 0=>9
00015      * A VALUE OF FF INDICATES A DEAD MAN
00016      * PROGRAM MAKES GREAT USE OF TRUG'S PLOT
00017      * ROUTINE SO WOULD NOT PRODUCE SATIS OUTPUT
00018      * ON A TELETYPE.
00019      *=====
00020      *
00021 0000 0001 MINAT RMB 1 MINOTAURS COORDS
00022 0001 0013 MEN RMB &19 20 LOCS RESERVED FOR
00023 0014 0001 ENDMEN RMB 1 MENS COORDS.
00024 0015 0001 MANCNT RMB 1 MAN-COUNT
00025 0016 0002 XTEMP RMB 2
00026 0018 0002 CKPTR RMB 2
00027 001A 0001 XINTMP RMB 1
00028 001B 0001 YINTMP RMB 1
00029 001C 0001 CNT3 RMB 1
00030 001D 0001 CNT5 RMB 1
00031 001E 0001 CNTLPS RMB 1
00032 001F 0001 RANCNT RMB 1
00033 0020 0004 RNDM RMB 4
00034 0024 0001 CNT4 RMB 1
00035 0030 ORG $0030
00036 0030 86 55 START LDA A #$55 PUT THE MINOTAUR ON THE
00037 0032 97 00 STA A MINAT MIDDLE OF THE BOARD
00038 0034 CE 0001 LDY #MEN ERASE ALL MEN
00039 0037 86 FF LDA A #$FF (FF = DEADMAN)
00040 0039 A7 00 NONSET STA A 0,X
00041 003B 08 INX
00042 003C 8C 0015 CPX #ENDMEN+1
00043 003F 26 F8 BNE NONSET
00044 0041 CE 0001 LDY #MEN
00045 0044 86 54 LDA A #$54 PUT 4 DUMMY MEN AT
00046 0046 97 14 STA A ENDMEN COORDINATES ADJACENT
00047 0048 86 65 LDA A #$65 TO MINOTAUR SO THESE
00048 004A 97 13 STA A ENDMEN-1 COORDS WONT BE CHOSEN
00049 004C 86 56 LDA A #$56 FOR THE RANDOMLY PLACED
00050 004E 97 12 STA A ENDMEN-2 MEN. THESE WILL BE
00051 0050 86 45 LDA A #$45 ERASED LATER.
00052 0052 97 11 STA A ENDMEN-3
00053 0054 BD 0313 GRAND JSR RANDOM GIVE FIRST TEN MEN
00054 0057 16 TAB RANDOM COORDINATES
00055 0058 84 F0 AND A #
00056 005A 81 90 CMP A #$90 IS COORDINATE IN RANGE ?
00057 005C 22 F6 BHI GRAND BRANCH IF NOT

```


00058	005E	C4	OF	AND B	#	
00059	0060	C1	09	CMP B	#\$09	IS Y COORDINATE IN RANGE
00060	0062	22	F0	BHI	GRAND	BRANCH IF NOT
00061	0064	1B		ABA		A NOW = X,Y
00062	0065	81	55	CMP A	#\$55	IS COORDINATE = MINOTAU
00063	0067	27	EB	BEQ	GRAND	BRANCH IF SO
00064	0069	BD	0276	JSR	CHEK	COORD ALREADY OCCU PI?
00065	006C	25	E6	BCS	GRAND	BRANCH IF SO
00066	006E	A7	00	STA A	0,X	COORDINATE OK SO STORE
00067	0070	08		INX		POINT X AT NEXT MAN
00068	0071	8C	000B	CPX	#\$000B	TEN MEN POSITIONED ?
00069	0074	26	DE	ENE	GRAND	IF NOT THEN DO NEXT
00070	0076	86	FF	LDA A	#\$FF	ERASE DUMMY MEN
00071	0078	97	14	STA A	ENDMEN	
00072	007A	97	13	STA A	ENDMEN-1	
00073	007C	97	12	STA A	ENDMEN-2	
00074	007E	97	11	STA A	ENDMEN-3	
00075	0080	BD	028E	JSR	SCREEN	DISPLAY MEN AND BOARD
00076	0083	BD	037C	JSR	PLOTM	DISPLAY MINOTAUR
00077	0086	86	02	LDA A	#\$02	
00078	0088	97	24	STA A	CNT4	
00079	008A	CE	0495	GETRAT LDX	#RATE	GET RATING
00080	008D	BD	0364	JSR	PRTREP	
00081	0090	BD	E1AC	JSR	INCH	
00082	0093	81	4D	CMP A	#'M	MASTER ?
00083	0095	27	07	BEQ	CONT	
00084	0097	81	42	CMP A	#'B	BEGINNER ?
00085	0099	26	EF	BNE	GETRAT	NEITHER TRY AGAIN
00086	009B	7C	0024	INC	CNT4	
00087	009E	86	0A	CONT LDA A	#\$10	SET MAN COUNT TO TEN
00088	00A0	97	15	STA A	MANCNT	
00089	00A2	86	03	LDA A	#3	SET CNTLPS TO THREE
00090	00A4	97	1E	STA A	CNTLPS	
00091	00A6	86	05	YRSET LDA A	#\$05	SET YEAR COUNT
00092	00A8	97	1D	STA A	CNT5	TOTAL TIME=CNTLPS * CNT5
00093				*		
00094	00AA	96	24	MOVSET LDA A	CNT4	SET CNT3=CNT4 MAN MOVES
00095	00AC	97	1C	STA A	CNT3	PER MINOTAUR MOVE
00096	00AE	CE	0488	GETMOV LDX	#MOVE	PLAYER TO MAKE A MOVE
00097	00B1	BD	0364	JSR	PRTREP	
00098	00B4	8D	1E	BSR	NUMIN	GET X COORDINATE
00099	00B6	16		TAB		
00100	00B7	86	2C	LDA A	#',	OUTPUT A COMMA
00101	00B9	BD	E1D1	JSR	CUTCH	
00102	00BC	D7	1A	STA B	XINTMP	SAVE X COORD
00103	00BE	58		ASL B		ROTATE X COORD TO MS 4
00104	00BF	58		ASL B		BITS OF WORD
00105	00C0	58		ASL B		
00106	00C1	58		ASL B		

00107	00C2	8D	10	BSR	NUMIN	GET Y COORD
00108	00C4	97	1B	STA A	YINTMP	SAVE Y COORD
00109	00C6	1B		ABA		A = X,Y
00110	00C7	BD	0276	JSR	CHEK	MAN THERE TO MOVE?
00111	00CA	25	31	BCS	MANAT	BRANCH IF SO
00112	00CC	CE	039C	LDX	#NOMAN	TELL THE PLAYER THAT
00113	00CF	BD	036C	JSR	PRINT	THERES NO MAN THERE
00114	00D2	20	DA	BRA	GETMOV	
00115	00D4	BD	E1AC	JSR	INCH	INPUT HALF A COORD AND
00116	00D7	81	47	CMP A	#'G	
00117	00D9	27	0F	BEQ	GIVEUP	
00118	00DB	81	30	CMP A	#'0	CHECK THAT IT IS NUMERIC
00119	00DD	25	07	BCS	GTM	CLR MS 4 BITS AND RTS
00120	00DF	81	39	CMP A	#'9	IF IT IS NORMALISE STACK
00121	00E1	22	03	BHI	GTM	BRANCH TO GETMOV IF NOT
00122	00E3	84	0F	AND A	#	
00123	00E5	39		RTS		HAS GIVEN UP SO
00124	00E6	31		INS		PRINT BOARD AND END
00125	00E7	31		INS		
00126	00E8	20	C4	BRA	GETMOV	
00127	00EA	CE	04A8	LDX	#GIVUP	
00128	00ED	BD	0359	JSR	PDTDEL	
00129	00F0	31		INS		
00130	00F1	31		INS		
00131	00F2	7E	0258	JMP	PRBEND	
00132	00F5	CE	03CD	LDX	#TOFRST	TELL HIM THAT HE CAN'T
00133	00F8	ED	036C	JSR	PRINT	MOVE THAT FAR
00134	00FB	20	B1	BRA	GETMOV	
00135	00FD	CE	040A	LDX	#TOSTR	PRINT 'TO ? '
00136	0100	BD	E07E	JSR	PDATA	
00137	0103	8D	CF	BSR	NUMIN	GET X DEST COORD
00138	0105	16		TAB		
00139	0106	C0	04	SUB B	#\$04	CHECK THAT MOVE IS
00140	0108	D1	1A	CMP B	XINTMP	NOT TOO BIG
00141	010A	2A	E9	BPL	TOOFAR	GOTO TOOFAR IF IT IS.
00142	010C	CB	07	ADD B	#\$07	
00143	010E	D1	1A	CMP B	XINTMP	
00144	0110	2B	E3	BMI	TOOFAR	
00145	0112	48		ASL A		DEST COORD IS OK
00146	0113	48		ASL A		SHIFT IT TO MS BITS
00147	0114	48		ASL A		
00148	0115	48		ASL A		
00149	0116	97	1A	STA A	XINTMP	
00150	0118	86	2C	LDA A	#',	
00151	011A	BD	E1D1	JSR	OUTCH	
00152	011D	8D	B5	BSR	NUMIN	GET Y DEST COORD
00153	011F	BD	0374	JSR	CLCSR	CLEAR CURSOR
00154	0122	16		TAB		
00155	0123	C0	04	SUB B	#4	CHECK THAT MOVE IS NOT

00156 0125 D1 1B	CMP B	YINTMP	TOO BIG
00157 0127 2A CC	BPL	TOOFAR	
00158 0129 CB 07	ADD B	#\$07	
00159 012B D1 1B	CMP B	YINTMP	
00160 012D 2B C6	BMI	TOOFAR	
00161 012F 9B 1A	ADD A	XINTMP	A = X,Y
00162 0131 BD 02F2	JSR	CHEK2	CHECK THAT DEST IS VAC
00163 0134 24 08	BCC	OK	BRANCH IF OK
UPIED 0136 CE 03E1	LDX	#ALRDY	TELL PLAYER COORD IS OCC
00165 0139 BD 036C	JSR	PRINT	
00166 013C 20 BD	BRA	GTMLVK	
00167 013E DE 18 OK	LDX	CKPTR	CKPTR IS MAN TO MOVE
00168 0140 A7 00	STA A	0,X	STORE NEW COORDS
00169 0142 7A 001C	DEC	CNT3	MINOTAURS TURN TO MOVE ?
00170 0145 26 B4	BNE	GTMLVK	BRANCH IF NOT
00171	*		
00172	*+++	CAN THE MINOTAUR EAT ?	
00173	*+++	MINOTAUR EATS IN DIAGON*AL DIRECTION	
00174	*		
00175 0147 96 00	LDA A	MINAT	
00176 0149 8B 11	ADD A	#\$11	EAT DOWN RIGHT ?
00177 014B BD 0276	JSR	CHEK	CHECKS FOR A MAN AT
00178 014E 25 3C	BCS	YUMYUM	COORDS IN A.
00179 0150 80 22	SUB A	#\$22	EAT UP LEFT ?
00180 0152 BD 0276	JSR	CHEK	
00181 0155 25 35	BCS	YUMYUM	
00182 0157 8B 02	ADD A	#\$02	EAT DOWN LEFT ?
00183 0159 BD 0276	JSR	CHEK	
00184 015C 25 2E	BCS	YUMYUM	
00185 015E 8B 20	ADD A	#\$20	EAT UP RIGHT ?
00186 0160 80 02	SUB A	#\$02	
00187 0162 BD 0276	JSR	CHEK	
00188 0165 25 25	BCS	YUMYUM	
00189	*		
00190	*+++	CAN THE MINOTAUR MOVE ?	
00191	*+++	MINOTAUR ONE PLACE UP,DWN LFT OR RIGHT	
00192	*		
00193 0167 7F 001F	CLR	RANCNT	RANCNT IS DIRECTI ON TRI
00194 016A D6 00	MOVMIN LDA B	MINAT	
00195 016C 7C 001F	INC	RANCNT	RANCNT-FF IF ALL FOUR
00196 016F 27 40	BEQ	CANTMV	HAVE BEEN TRIED
00197 0171 7A 001F	DEC	RANCNT	
00198 0174 BD 0313	JSR	RANDOM	GET MOVE DIRECTION
00199 0177 84 03	AND A	#	
00200 0179 26 3F	BNE	NOTO	
00201 017B 86 11	LDA A	#\$11	UPDATE RANCNT
00202 017D 9A 1F	ORA A	RANCNT	
00203 017F 97 1F	STA A	RANCNT	
00204 0181 C0 01	SUB B	#1	MOVE MIN UP ?

00205	0183	17		TBA	
00206	0184	84	OF	AND A	#\$0F OFF BOARD ?
00207	0186	81	OF	CMP A	#\$0F
00208	0188	27	E0	BEQ	MOVMIN
NT209	018A	20	60	BRA	MOVCK :CHECKS POSITION IS VACA
00210	018C	7A	001D	YUMYUM DEC	CNT5 MINOTAUR EATS A MAN
0N211	018F	97	00	STA A	MINAT UPDATE MINOTAUR RS POSITI
00212	0191	BD	0332	JSR	INCLIN
00213	0194	7C	001D	YMNINC INC	CNT5
00214	0197	CE	04BA	LDX	YUM TELL MINOTAUR HAS EATE
00215	019A	BD	0359	JSR	PDTDEL
00216	019D	86	FF	LDA A	#\$FF -ERASE MAN (FF=DEADMAN)
00217	019F	DE	18	LDX	CKPTR
00218	01A1	A7	00	STA A	0,X
00219	01A3	7A	0015	DEC	MANCNT ALL MEN GONE ?
00220	01A6	26	4C	BNE	DISPLY DISPLAY: DISPLAYS BOARD
00221	01A8	CE	0411	LDX	#NOMEN TELL PLAYER ALL MEN
00222	01AB	BD	0359	JSR	PDTDEL ARE GONE
00223	01AE	7E	0258	JMP	PRBEND PRINT BOARD AND END
00224	01B1	CE	03E5	CANTMV LDX	#WINSTR MINOTAUR SURROUNDED
00225	01B4	BD	0359	JSR	PDTDEL TELL AND PAT HIS BACK
00226	01B7	7E	0258	JMP	PRBEND
00227	01BA	4A		NOTO DEC A	
00228	01BB	26	0E	BNE	NOT1
00229	01BD	86	22	LDA A	#\$22 UPDATE RANCNT
00230	01BF	9A	1F	ORA A	RANCNT
00231	01C1	97	1F	STA A	RANCNT
00232	01C3	CB	10	ADD B	#\$10 MOVE MIN RIGHT ?
00233	01C5	C1	9F	CMP B	#\$9F OFF BOARD ?
00234	01C7	2A	A1	BPL	MOVMIN
00235	01C9	20	21	BRA	MOVCK CHECK IF POSITION VACANT
00236	01CB	4A		NOT1 DEC A	
00237	01CC	26	11	BNE	NOT2
00238	01CE	86	44	LDA A	#\$44 UPDATE RANCNT
00239	01D0	9A	1F	ORA A	RANCNT
00240	01D2	97	1F	STA A	RANCNT
00241	01D4	CB	01	ADD B	#\$01 MOVE MIN DOWN ?
00242	01D6	17		TBA	
00243	01D7	84	0F	AND A	#\$0F
00244	01D9	81	0A	CMP A	#\$0A
00245	01DB	27	8D	BEQ	MOVMIN
00246	01DD	20	0D	BRA	MOVCK CHECK IF VACANT
00247	01DF	86	88	NOT2 LDA A	#\$88
00248	01E1	9A	1F	ORA A	RANCNT
00249	01E3	97	1F	STA A	RANCNT
00250	01E5	C0	10	SUB B	#\$10 MOVE MIN LEFT ?
00251	01E7	24	03	BCC	MOVCK
00252	01E9	7E	016A	MVMNLK JMP	MOVMIN
00253				*	

00254	01EC	17	MOVCK	TBA	
00255	01ED	BD 02F2	JSR	CHEK2	IF POSN OCCUPIED RETURN
00256	01FO	25 F7	BCS	MVMNLK	TO MOVMIN
00257	01F2	D7 00	STA B	MINAT	IF NOT UPDATE MINS COORD
00258	01F4	BD 028E	DISPLY JSR	SCREEN	DISPLAY BOARD AND MEN
00259	01F7	7A 001D	DEC	CNT5	TIME FOR A NEW MAN ?
00260	01FA	27 03	BEQ	NEWMAN	
00261	01FC	7E 00AA	JMP	MOVSET	
00262	01FF	7A 001E	NEWMAN DEC	CNTLPS	
00263	0202	27 4E	BEQ	NOTIM	
00264	0204	CE 0432	LDX	#NEWONE	TEL PLAYER HES GOT
00265	0207	BD 0364	JSR	PRTREP	A NEW MAN
00266	020A	8D 24	BSR	INNEW	GET X COORD FOR POSN
00267	020C	16	TAB		OF NEWMAN FROM PLAYER
00268	020D	86 2C	LDA A	#',	
00269	020F	BD E1D1	JSR	OUTCH	
00270	0212	8D 1C	BSR	INNEW	GET Y COORD
00271	0214	BD 0374	JSR	CLCSR	CLEAR CURSOR.
00272	0217	58	ASL B		
00273	0218	58	ASL B		
00274	0219	58	ASL B		
00275	021A	58	ASL B		
00276	021B	1B	ABA		
00277	021C	91 00	CMP A	MINAT	ON MINOTAUR ?
00278	021E	26 03	BNE	CONT1	
00279	0220	7E 0194	JMP	YMNINC	:EATS NEW MAN
00280	0223	BD 0276	CONT1 JSR	CHEK	POSN VACANT ?
00281	0226	24 1A	BCC	NEWPUT	
00282	0228	CE 03B1	LDX	#ALRDY	POSN NOT VACANT
00283	022B	BD 036C	JSR	PRINT	
00284	022E	20 CF	BRA	NEWMAN	
00285	0230	BD E1AC	INNEW JSR	INCH	GET A COORD FOR NEWMAN
00286	0233	81 30	CMP A	#'0	BRANCH TO NEWMAN IF
00287	0235	25 07	BCS	GTMM	INPUT IS NOT NUMERIC
00288	0237	81 39	CMP A	#'9	CLR 4 MS BITS AND
00289	0239	22 03	BHI	GTMM	RTS IF NUMERIC
00290	023B	84 0F	AND A	#	
00291	023D	39	RTS		
00292	023E	31	GTMM INS		
00293	023F	31	INS		
00294	0240	20 BD	BRA	NEWMAN	
00295	0242	16	NEWPUT TAB		
00296	0243	86 FF	LDA A	#\$FF	FIND NEXT DEADMAN
00297	0245	BD 0276	JSR	CHEK	
00298	0248	DE 18	LDX	CKPTR	PUT NEWMAN IN HIS PLACE
00299	024A	E7 00	STA B	0,X	
00300	024C	7C 0015	INC	MANCNT	
00301	024F	7E 00A6	JMP	YRSET	
00302	0252	CE 0443	NOTIM LDX	#NTIMST	TIMES UP !

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00303 0255 BD 0359      JSR   PDTDEL
00304                  *
00305 0258 BD 028E PRBEND JSR   SCREEN   DISPLAY BOARD AND MEN
00306 025B BD 037C      JSR   PLOTM    DISPLAY MINOTAUR
00307 025E CE 045B      LDX   #AGAIN   PLAY AGAIN?
00308 0261 BD 0364      JSR   PRTREP
00309 0264 BD E1AC      JSR   INCH
00310 0267 81 59        CMP   A   #'Y
00311 0269 26 03        BNE   TRET
00312 026B 7E 0030      JMP   START
00313 026E 86 17 TRET   LDA   A   #$17
00314 0270 B7 A083      STA   A   ROWPTR   WRITE ON BOTTOM LINE
00315 0273 7E E0E3      JMP   TBUG    AGAIN FOR TBUG
00316                  *
00317                  *   SUBROUTINES AND
00318                  *   TEXT CONSTANTS.
00319                  *
00320                  *+++   CHEK
00321 0276 DF 16          CHEK  STX   XTEMP   CARRY IF MAN AT COORDS
00322 0278 CE 0001      LDX   #MEN     IN A FOUND CKPTR=ADDRESS
IS323 027B A1 00      CK    CMP   A   0,X   OF MAN OTHER WISE CARRY
00324 027D 27 09      BEQ   SETRET    IS CLEARED
00325 027F 08          INX
00326 0280 8C 0015      CPX   #ENDMEN+1
00327 0283 26 F6        BNE   CK
00328 0285 0C          CLC
00329 0286 20 01        BRA   NOTSET
00330 0288 0D          SETRET SEC
00331 0289 DF 18      NOTSET STX   CKPTR
00332 028B DE 16      LDX   XTEMP
00333 028D 39          RTS
00334                  **+++   SCREEN
00335 028E CE F800 SCREEN  LDX   #VDU     DISPLAYS BOARD AND MEN
00336 0291 86 20        LDA   A   #'
00337 0293 A7 00      CLR   STA   A   0,X
00338 0295 08          INX
00339 0296 8C FBF8      CPX   #VDUEND
00340 0299 26 F8        BNE   CLR
00341 029B 86 02        LDA   A   #$2     PRINT 'MINOTAUR' AND X
00342 029D B7 A083      STA   A   ROWPTR   NOS. 0 => 9
00343 02A0 CE 0469      LDX   #MINSTR
00344 02A3 BD E07E      JSR   PDATA
00345 02A6 CE A080      LDX   #SCRACH
00346 02A9 6C 02        INC   2,X
00347 02AB 6C 02        INC   2,X
00348 02AD 6C 03        INC   3,X
00349 02AF 86 30        LDA   A   #'0
00350 02B1 BD E220 YGRID JSR   PLOT     TYPE Y GRID NOS. 0 => 9
00351 02B4 17          TBA

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00352	02B5	6C	03	INC	3,X	
00353	02B7	4C		INC	A	
00354	02B8	81	3A	CMP	A	#\$3A
00355	02BA	26	F5	BNE		YGRID
00356	02BC	CE	0001	LDX		#MEN
00357	02BF	A6	00	DISMAN	LDA	A
00358	02C1	81	FF		CMP	A
00359	02C3	27	1D		BEQ	
00360	02C5	16			TAB	
00361	02C6	DF	16		STX	XTEMP
00362	02C8	CE	A080		LDX	#SCRACH
00363	02CB	84	FD		AND	A
00364	02CD	44			LSR	A
00365	02CE	44			LSR	A
00366	02CF	44			LSR	A
00367	02D0	44			LSR	A
00368	02D1	8B	03		ADD	A
00369	02D3	A7	02		STA	A
00370	02D5	C4	0F		AND	B
00371	02D7	CB	03		ADD	B
00372	02D9	E7	03		STA	B
00373	02DB	86	2A		LDA	A
00374	02DD	BD	E220		JSR	PLOT
00375	02E0	DE	16		LDX	XTEMP
00376	02E2	08		SKIP	INX	
00377	02E3	8C	0015		CPX	#ENDMEN+1
00378	02E6	26	D7		BNE	DISMAN
00379	02E8	CE	A080		LDX	#SCRACH
00380	02EB	6F	02		CLR	2,X
00381	02ED	86	0F		LDA	A
00382	02EF	A7	03		STA	A
00383	02F1	39			RTS	
00384				*+++	CHEK	2
00385	02F2	91	00	CHEK2	CMP	A
00386	02F4	26	0B		BNE	CHEK3
ME387	02F6	31			INS	
00388	02F7	31			INS	
00389	02F8	DE	18		LDX	CKPTR
00390	02FA	86	FF		LDA	A
00391	02FC	A7	00		STA	A
00392	02FE	7E	0194		JMP	YMNINC
00393	0301	CE	0001	CHEK3	LDX	#MEN
00394	0304	A1	00	COMP	CMP	A
00395	0306	27	09		BEQ	SET
00396	0308	08			INX	
00397	0309	8C	0015		CPX	#ENDMEN+1
00398	030C	26	F6		BNE	COMP
00399	030E	0C			CLC	
00400	030F	20	01		BRA	CLRD

DISPLAY MEN ON BOARD AS
 ASTERISKS
 DEADMAN ?
 DONT DISPLAY DEADMAN
 FROM MAN COORDS FORM
 #
 FOR SCREEN
 PLOT '*' FOR MAN
 POINT X AT NEXT MAN
 END OF MEN ?
 INIT ROWPTR TO WRITE
 ON 16TH LINE
 KILLS MAN IF = MINOTAU
 CKPTR IS NOT AFFECTED
 OTHERWISE FUNCTION IS SA
 AS CHEK 1

00401	0311	0D	SET	SEC	
00402	0312	39	CLRD	RTS	
00403			*+++*	RANDOM	
00404	0313	37	RANDOM	PSH B	:PUTS RANDOM NO IN A
00405	0314	C6 08		LDA B	#8 USED TO DECIDE MINOTAURS
00406	0316	96 23	RPT	LDA A	RNDM+3 MOVE DIRECTION.
00407	0318	48		ASL A	AND TO PLACE THE MEN ON
00408	0319	48		ASL A	THE BOARD
00409	031A	48		ASL A	
00410	031B	98 23		EOR A	RNDM+3
00411	031D	48		ASL A	
00412	031E	48		ASL A	
00413	031F	79 0020		ROL	RNDM
00414	0322	79 0021		ROL	RNDM+1
00415	0325	79 0022		ROL	RNDM+2
00416	0328	79 0023		ROL	RNDM+3
00417	032B	5A		DEC B	
00418	032C	26 B8		BNE	RPT
00419	032E	33		PUL B	
00420	032F	96 20		LDA A	RNDM
00421	0331	39		RTS	
00422			*+++*	INCLIN	
00423	0332	B6 A083	INCLIN	LDA A	ROWPTR WRITE ON NEXT LINE
00424	0335	4C		INC A	WITHOUT SCROLL SO AS
00425	0336	81 17		CMP A	#23 NOT TO DISTURB BOARD
00426	0338	26 02		BNE	LIN1 CLEARS LINE TO BE WRIT
00427	033A	86 0F		LDA A	#15 ON AND LINE BELOW
00428	033C	B7 A083	LIN1	STA A	ROWPTR AFTER LINE 24 ROUTINE
00429	033F	8D 09		BSR	SPSOUT WRITE ON LINE 16
00430	0341	7C A083		INC	ROWPTR
00431	0344	8D 04		BSR	SPSOUT
00432	0346	B7 A083		STA A	ROWPTR
00433	0349	39		RTS	
00434			*+++*	SPSOUT	
00435	034A	36	SPSOUT	PSH A	:OUTPUTS 38 SPACES
00436	034B	C6 26		LDA B	#38 FOR CLEARING LINES
00437	034D	8D 25		BSR	CLCSR CLEAR CURSOR
00438	034F	BD B0CC	OUTSP	JSR	OUTS
00439	0352	5A		DEC B	
00440	0353	26 FA		BNE	OUTSP
00441	0355	8D 1D		BSR	CLCSR CLEAR CURSOR
00442	0357	32		PUL A	
00443	0358	39		RTS	
00444			*+++*	PDTDEL	
00445	0359	8D 11	PDTDEL	BSR	PRINT :PRINT STRING AND THEN
00446	035B	86 04		LDA A	#34 WAIT FOR APPROX 2 SECS.
00447	035D	09	DEX	DEX	
00448	035E	26 FD		BNE	DEX
00449	0360	4A		DEC A	


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00450 0361 26 FA      BNE    DEX
00451 0363 39      RTS
00452      *+++    PRTRP
00453 0364 8D 0E      PRTRP BSR    CLCSR
00454 0366 8D CA      BSR    INCLIN
00455 0368 BD E07E      JSR    PDATA
00456 036B 39      RTS
00457      *+++    PRINT
00458 036C 8D C4      PRINT BSR    INCLIN    WRITE STRING ON NEXT
00459 036E BD E07E      JSR    PDATA    LINE WITHOUT SCROLL
00460 0371 8D 01      BSR    CLCSR
00461 0373 39      RTS
00462      *+++    CLCSR
00463 0374 36      CLCSR PSH A      OUTPUT CARRIAGE RETURN
00464 0375 86 0D      LDA A    #$0D    AND SO CLEAR CURSOR
00465 0377 BD E1D1      JSR    OUTCH
00466 037A 32      PUL A
00467 037B 39      RTS
00468      *+++    PLOTM
00469 037C CE A080      PLOTM LDX    #SCRACH    TYPE MINOTAUR ON BOARD
00470 037F 96 00      LDA A    MINAT    FORM PLOT COORDS FROM
00471 0381 16      TAB      MINOTAURS COORDS
00472 0382 84 F0      AND A    #
00473 0384 C4 0F      AND B    #
00474 0386 CB 03      ADD B    #3
00475 0388 E7 03      STA B    3,X
00476 038A 44      LSR A
00477 038B 44      LSR A
00478 038C 44      LSR A
00479 038D 44      LSR A
00480 038E 8B 03      ADD A    #3
00481 0390 A7 02      STA A    2,X
00482 0392 86 4D      LDA A    #'M    'M' = MINOTAUR
00483 0394 BD E220      JSR    PLOT    PLOT IT
00484 0397 86 0F      LDA A    #&15    WRITE ON LINE 16 AGAIN
00485 0399 A7 03      STA A    3,X
00486 039B 39      RTS
00487      *+++++*
00488      *    TEXT CONSTANTS    *
00489      *+++++*
00490 039C 54      NOMAN FCC    /THERES NO MAN THERE /
      039D 48
      039E 45
      039F 52
      03A0 45
      03A1 53
      03A2 20
      03A3 4E
      03A4 4F

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	03A5	20			
	03A6	4D			
	03A7	41			
	03A8	4E			
	03A9	20			
	03AA	54			
	03AB	48			
	03AC	45			
	03AD	52			
	03AE	45			
	03AF	20			
00491	03B0	04		FCB	\$04
00492	03B1	54	ALRDY	FCC	/THERES A MAN THERE ALREADY./
	03B2	48			
	03B3	45			
	03B4	52			
	03B5	45			
	03B6	53			
	03B7	20			
	03B8	41			
	03B9	20			
	03BA	4D			
	03BB	41			
	03BC	4E			
	03BD	20			
	03BE	54			
	03BF	48			
	03C0	45			
	03C1	52			
	03C2	45			
	03C3	20			
	03C4	41			
	03C5	4C			
	03C6	52			
	03C7	45			
	03C8	41			
	03C9	44			
	03CA	59			
	03CB	2E			
00493	03CC	04		FCB	\$04
00494	03CD	59	TOFRST	FCC	/YOU CANT MOVE THAT FARI/
	03CE	4F			
	03CF	55			
	03D0	20			
	03D1	43			
	03D2	41			
	03D3	4E			
	03D4	54			
	03D5	20			

	03D6	4D		
	03D7	4F		
	03D8	56		
	03D9	45		
	03DA	20		
	03DB	54		
	03DC	48		
	03DD	41		
	03DE	54		
	03DF	20		
	03E0	46		
	03E1	41		
	03E2	52		
	03E3	21		
00495	03E4	04	FCB	\$04
00496	03E5	54	WINSTR FCC	/THE MINOTAUR IS SURROUNDED. /
	03E6	48	FCC	/YOU WIN! /
	03E7	45		
	03E8	20		
	03E9	4D		
	03EA	49		
	03EB	4E		
	03EC	4F		
	03ED	54		
	03EE	41		
	03EF	55		
	03F0	52		
	03F1	20		
	03F2	49		
	03F3	53		
	03F4	20		
	03F5	53		
	03F6	55		
	03F7	52		
	03F8	52		
	03F9	4F		
	03FA	55		
	03FB	4E		
	03FC	44		
	03FD	45		
	03FE	44		
	03FF	2E		
	0400	20		
	0401	59		
	0402	4F		
	0403	55		
	0404	20		
	0405	57		
	0406	49		

	0407	4E			
	0408	21			
00497	0409	04		FCB	\$04
00498	040A	20	TOSTR	FCC	/ TO ? /
	040B	54			
	040C	4F			
	040D	20			
	040E	3F			
	040F	20			
00499	0410	04		FCB	\$04
00500	0411	41	NOMEN	FCC	/ALL OF YOUR MEN HAVE BEEN EATEN! /
	0412	4C			
	0413	4C			
	0414	20			
	0415	4F			
	0416	46			
	0417	20			
	0418	59			
	0419	4F			
	041A	55			
	041B	52			
	041C	20			
	041D	4D			
	041E	45			
	041F	4E			
	0420	20			
	0421	48			
	0422	41			
	0423	56			
	0424	45			
	0425	20			
	0426	42			
	0427	45			
	0428	45			
	0429	4E			
	042A	20			
	042B	45			
	042C	41			
	042D	54			
	042E	45			
	042F	4E			
	0430	21			
00501	0431	04		FCB	\$04
00502	0432	4E	NEWONE	FCC	/NEW MAN WHERE ? /
	0433	45			
	0434	57			
	0435	20			
	0436	4D			
	0437	41			

	0438	4E			
	0439	20			
	043A	57			
	043B	48			
	043C	45			
	043D	52			
	043E	45			
	043F	20			
	0440	3F			
	0441	20			
00503	0442	04		FCB	\$04
00504	0443	59	NTIMST	FCC	/YOU'VE RUN OUT OF TIME! /
	0444	4F			
	0445	55			
	0446	27			
	0447	56			
	0448	45			
	0449	20			
	044A	52			
	044B	55			
	044C	4E			
	044D	20			
	044E	4F			
	044F	55			
	0450	54			
	0451	20			
	0452	4F			
	0453	46			
	0454	20			
	0455	54			
	0456	49			
	0457	4D			
	0458	45			
	0459	21			
00505	045A	04		FCB	\$04
00506	045B	50	AGAIN	FCC	/PLAY AGAIN ? /
	045C	4C			
	045D	41			
	045E	59			
	045F	20			
	0460	41			
	0461	47			
	0462	41			
	0463	49			
	0464	4E			
	0465	20			
	0466	3F			
	0467	20			
00507	0468	04		FCB	\$04

00508	0469	0D	MINSTR	FCB	\$0D
00509	046A	20		FCC	/ MINOTAUR/
	046B	20			
	046C	20			
	046D	20			
	046E	4D			
	046F	49			
	0470	4E			
	0471	4F			
	0472	54			
	0473	41			
	0474	55			
	0475	52			
00510	0476	0D		FCB	\$D,\$A,\$A
	0477	0A			
	0478	0A			
00511	0479	20		FCC	/ 0123456789/
	047A	20			
	047B	20			
	047C	30			
	047D	31			
	047E	32			
	047F	33			
	0480	34			
	0481	35			
	0482	36			
	0483	37			
	0484	38			
	0485	39			
00512	0486	0D		FCB	\$D,\$04
	0487	04			
00513	0488	4D	MOVE	FCC	/MOVE FROM ? /
	0489	4F			
	048A	56			
	048B	45			
	048C	20			
	048D	46			
	048E	52			
	048F	4F			
	0490	4D			
	0491	20			
	0492	3F			
	0493	20			
	0464	4E			
	0465	20			
	0466	3F			
	0467	20			
00507	0468	04		FCB	\$04
00508	0469	0D	MINSTR	FCB	\$0D

00509	046A 20	FCC	/	MINOTAUR/
	046B 20			
	046C 20			
	046D 20			
	046E 4D			
	046F 49			
	0470 4E			
	0471 4F			
	0472 54			
	0473 41			
	0474 55			
	0475 52			
00510	0476 0D	FCB		\$D,\$A,\$A
	0477 0A			
	0478 0A			
00511	0479 20	FCC	/	0123456789/
	047A 20			
	047B 20			
	047C 30			
	047D 31			
	047E 32			
	047F 33			
	0480 34			
	0481 35			
	0482 36			
	0483 37			
	0484 38			
	0485 39			
00512	0486 0D	FCB		\$D,\$04
	0487 04			
00513	0488 4D	MOVE FCC		/MOVE FROM ? /
	0489 4F			
	048A 56			
	048B 45			
	048C 20			
	048D 46			
	048E 52			
	048F 4F			
	0490 4D			
	0491 20			
	0492 3F			
	0493 20			
00514	0494 04	FCB		\$04
00515	0495 52	RATE FCC		/RATING (B OR M) ? /
	0496 41			
	0497 54			
	0498 49			
	0499 4E			
	049A 47			

	049B	20			
	049C	28			
	049D	42			
	049E	20			
	049F	4F			
	04A0	52			
	04A1	20			
	04A2	4D			
	04A3	29			
	04A4	20			
	04A5	3F			
	04A6	20			
00516	04A7	04		FCB	\$04
00517	04A8	59	GIVUP	FCC	/YOU'VE GIVEN UP I/
	04A9	4F			
	04AA	55			
	04AB	27			
	04AC	56			
	04AD	45			
	04AE	20			
	04AF	47			
	04B0	49			
	04B1	56			
	04B2	45			
	04B3	4E			
	04B4	20			
	04B5	55			
	04B6	50			
	04B7	20			
	04B8	21			
00518	04B9	04		FCB	\$04
00519	04BA	54	YUM	FCC	/THE MINOTAUR HAS JUST /
	04BB	48			
	04BC	45			
	04BD	20			
	04BE	4D			
	04BF	49			
	04C0	4E			
	04C1	4F			
	04C2	54			
	04C3	41			
	04C4	55			
	04C5	52			
	04C6	20			
	04C7	48			
	04C9	53			
	04CA	20			
	04CB	4A			
	04CC	55			


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04CD 53

04CE 54
04CF 20
00520 04D0 48      FCC      /HAD A GOOD MEAL!/
04D1 41
04D2 44
04D3 20
04D4 41
04D5 20
04D6 47
04D7 4F
04D8 4F
04D9 44
04DA 20
04DB 4D
04DC 45
04DD 41
04DE 4C
04DF 21
00521 04E0 04      END      FCB      $04
00522      A080      SCRACH EQU      $A080
00523      A082      COLPTR EQU      $A082
00524      A083      ROWPTR EQU      $A083
00525      E07E      PDATA EQU      $E07E
00526      E0CC      OUTS EQU      $E0CC
00527      E0E3      TBUG EQU      $E0E3
00528      E1AC      INCH EQU      $E1AC
00529      E1D1      OUTCH EQU      $E1D1
00530      E220      PLOT EQU      $E220
00531      F800      VDU EQU      $F800
00532      FBFB      VDUEND EQU      $FBFB
00533                      END

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SYMBOL TABLE

MINAT	0000	MEN	0001	ENDMEN	0014	MANCNT	0015	XTEMP	0016
CKPTR	0018	XINTMP	001A	YINTMP	001B	CNT3	001C	CNT5	001D
CNTLPS	001E	RANCNT	001F	RNDM	0020	CNT4	0024	START	0030
NONSET	0039	GRAND	0054	GETRAT	008A	CONT	009E	YRSET	00A6
MOVSET	00AA	GETMOV	00AE	NUMIN	00D4	GTM	00E6	GIVEUP	00EA
TOOFAR	00F5	GTMVLK	00FB	MANAT	00FD	OK	013E	MOVMIN	016A
YUMYUM	018C	YMNINC	0194	CANTMV	01B1	NOTO	01BA	NOT1	01CB
NOT2	01DF	MVMNLK	01E9	MOVCK	01EC	DISPLY	01F4	NEWMAN	01FF
CONT1	0223	INNEW	0230	GTMM	023E	NEWPUT	0242	NOTIM	0252
PRBEND	0258	TRET	026E	CHEK	0276	CK	027B	SETRET	0288
NOTSET	0289	SCREEN	028E	CLR	0293	YGRID	02B1	DISMAN	02BF
SKIP	02E2	CHEK2	02F2	CHEK3	0301	COMP	0304	SET	0311
CLRD	0312	RANDOM	0313	RPT	0316	INCLIN	0332	LIN1	033C

SPSOUT 034A	OUTSP 034F	PDTDEL 0359	DEX 035D	PRTREP 0364
PRINT 036C	CLCSR 0374	PLOTM 037C	NOMAN 039C	ALRDY 03B1
TOFRST 03CD	WINSTH 03E5	TOSTH 040A	NOMEN 0411	NEWONE 0432
NTIMST 0443	AGAIN 045B	MINSTR 0469	MOVE 0488	RATE 0495
GIVUP 04A8	YUM 04BA	END 04E0	SCRACH A080	COLPTR A082
ROWPTR A083	PDATA E07E	OUTS E0CC	TBUG E0E3	INCH E1AC
OUTCH E1D1	PLOT E220	VDU F800	VDUEND FBF8	

300600004844521B

S113003086559700CE000186FFA700088C00152680
 S1130040F8CE000186549714866597138656971246
 S113005086459711BD03131684F0819022F6C40FD0
 S1130060C10922F01B815527EBBD02762586A700C6
 S1130070088C000B26DE86FF9714971397129711AE
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 S11300F031317E0258CE03CDBD036C20B1CE040A4B
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 S1130140A7007A001C26B496008B11BD0276253CCC
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